Hormone administration / chemical alterations to endocrine state

For making oestrogens, progestogens or androgens

Preparation:
• Generally it is best to do these post-OVX/GDX of an animal to ensure gonadal steroids do not bias your hormone levels.
• Some labs will couple GDX with adrenalectomy to be certain, but I don't think that is horrifically necessary.
• Check the final concentration for what you need, rather than relying on the concentrations written above

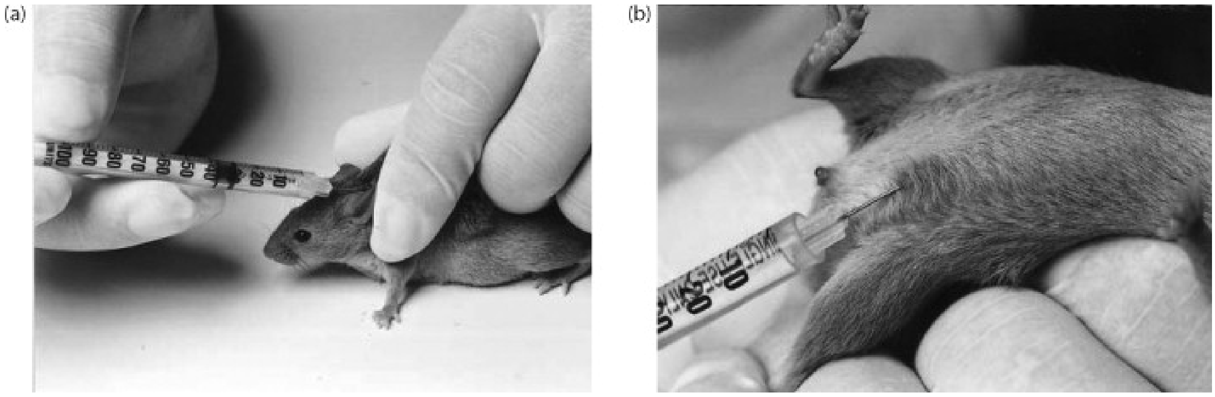
Oestrogens (~1 µg/inj.)	**Progestogens (~500 µg/inj)**	**Androgens (~250 µg/inj)**
17ß-oestradiol	Progesterone	Dihydrotestosterone
Oestradiol benzoate		Testosterone *(will be aromatised to oestradiol)*

Making hormones:
• Hormones should be initially prepared as a 10X stock solution in either EtOH or DMSO (depending on the SDS sheet).
-- For oestrogens, make up a 100X stock rather than a 10X.
• Then, take 100 µL of the 10X stock and dilute in 900 µL of sesame oil for your working stock.

Example:
• *For a final conc. of 500 µg/injection of progesterone, take 50 mg progesterone and dissolve in 1 mL DMSO.*
• *Take 100 µL of progesterone/DMSO and mix with 900 µL sesame oil.*

Storage:
• Store all drugs, oils, and preparations at room temperature in a cool, dark space.
• 10X stocks will last for 2 months
• Oils should be used within a week.
• Do not store oils in the fridge/freezer or they will turn into margarine.

Injection:
• Take 100 µL of your 1X oil solution and inject it into the scruff of the mouse from the nose end.
• Aim to inject as close to just behind between the ears as possible (see image a)



• The oil should be between the scapulae so movement causes the oil to dissipate.
• Repeated daily injections can alternate from scruff to inguinal skin (see image b).

LONGER TERM ALTERATIONS VIA SILASTIC IMPLANT

You will need:

• Silastic tubing: *Dow-Corning* 1 mm ID; 2.125 mm OD
• Silastic medical adhesive: *Dow-Corning* Silastic medical adhesive silicon Type A
• 10X hormone stock.
• Small Petri dish
• ~20G Needle
• 2x ~10cc Syringe

Method:

1. Back fill your syringe with adhesive to 1.8 mL
2. Push out 1.8 mL of glue into the Petri dish.
3. Add 200 µL of your 10X Stock onto the glue
4. Using a pipette tip mix together the glue and the stock solution until it becomes a shiny ball.
5. Back fill a second syringe with your hormone/glue ball
6. Put the needle on the syringe
7. Place the tip of the needle into the tubing
8. In one smooth motion (this will require a lot of force and effort and yelling), push the hormone/glue through into the tube.
9. Do not stop once you start, absolutely no air bubbles should be in the glue in the tube.
10. Leave your glue tubes covered to set (generally over night)

Implantation:

1. Cut the tubing to 1 cm per 20 g mouse
2. This will provide the equivalent of a 100 µL 1X oil solution injection.
3. Following OVX/GDX insert blunt scissors subcutaneously and blunt dissect the fascia to form a pouch up the back of the animal
4. Using long forceps, insert the tubing as far up towards the scapulae as possible
5. Close the wound
6. Hormones will release within 24 hours and last up to a week

7. *Generally I would suggest waiting to day 2 (assuming surgery is day 0) before using the animals to let them recover and hormones to be adequately released*